

Ex. No. : 10

Date:

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A PYTHON PROGRAM TO IMPLEMENT DIMENSIONALITY REDUCTION – PCA

Aim:

To implement **Principal Component Analysis (PCA)** for dimensionality reduction on the credit card dataset.

Requirements:

Jupyter Notebook with Python, creditcard.csv dataset, and required Python libraries.

PROGRAM:

Step 1: Import Required Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
```

Step 2: Load the Dataset

```
df = pd.read_csv('creditcard.csv')
print(df.head())
```



...	Time	V1	V2	V3	V4	V5	V6	V7
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941
5 rows × 31 columns								

Step 3: Separate features and target

```
X = df.drop('Class', axis=1)
y = df['Class']
```

Step 4: Feature Scaling

```
scaler = StandardScaler()  
X_scaled = scaler.fit_transform(X)
```

Step 5: Apply PCA

```
pca = PCA(n_components=2)  
X_pca = pca.fit_transform(X_scaled)
```

Step 6: Create DataFrame for visualization

```
pca_df = pd.DataFrame(data=X_pca, columns=['PC1', 'PC2'])  
pca_df['Class'] = y
```

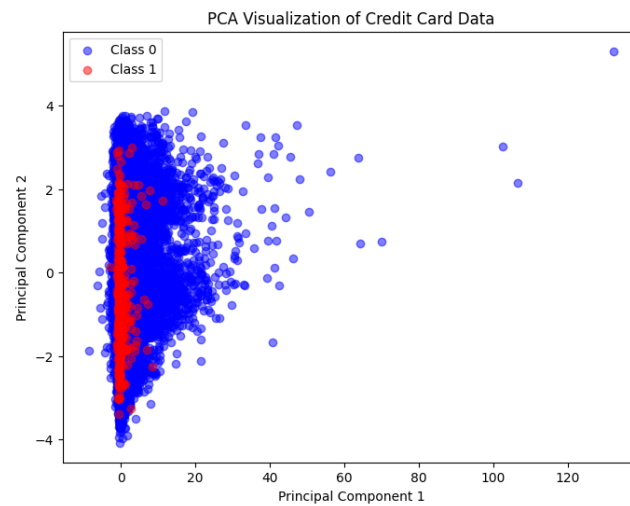
Step 7: Explained variance

```
print("Explained Variance Ratio:", pca.explained_variance_ratio_)
```

```
... Explained Variance Ratio: [0.06527521 0.05612313]
```

Step 8: Predict the Results

```
plt.figure(figsize=(8,6))  
colors = {0: 'blue', 1: 'red'}  
  
for cls in pca_df['Class'].unique():  
    subset = pca_df[pca_df['Class'] == cls]  
    plt.scatter(subset['PC1'], subset['PC2'],  
                c=colors[cls], label=f"Class {cls}", alpha=0.5)  
  
plt.xlabel('Principal Component 1')  
plt.ylabel('Principal Component 2')  
plt.title('PCA Visualization of Credit Card Data')  
plt.legend()  
plt.show()
```



Result:

Thus, PCA successfully reduced the dataset dimensions while retaining important variance, enabling better visualization and analysis.